

Task3_Architecture_Design - Soteria SOW Guardian

Option selected: Option B - Architecture Design

Entity: OCBC Bank Ltd

Jurisdictions: Singapore and United States

Domain: AI model risk management and AI governance for AML/KYC source-of-wealth due diligence

Tool name: Soteria SOW Guardian - Jurisdiction-Aware AI Governance Monitor for KYC Source-of-Wealth Review

1. Executive summary

Soteria SOW Guardian is a jurisdiction-aware governance tool for banks using AI in KYC source-of-wealth review. The tool is designed around OCBC Bank Ltd and Bank of Singapore. It does not replace a bank's source-of-wealth drafting assistant. Instead, it sits around the AI-assisted workflow and monitors whether the AI-generated narrative is supported by evidence, whether human reviewers exercise judgment, whether model behaviour is drifting, and whether the correct Singapore or US rule logic is active.

The architecture is built around the values position from Task 2: AI may assist compliance work, but it should not convert uncertainty into false confidence. The main risk in this domain is that an AI-generated source-of-wealth report may look coherent and professional even when the underlying evidence is weak. A memo or spreadsheet can document that review occurred, but it cannot continuously test claim-to-document evidence, detect contradictions, monitor rule versions, or identify human rubber-stamping.

The tool's core outputs are **Green**, **Amber**, **Red Friction**, **Amber + Governance Gap**, and **Red Friction + Governance Gap**. Red Friction blocks straight-through approval and requires senior compliance review.

2. Why the use case is realistic

OCBC Bank Ltd is a Singapore-headquartered bank with US agencies in New York and Los Angeles. OCBC's USA page states that its New York Agency and Los Angeles Agency are branches of OCBC Bank, and that the New York Agency is supervised by the New York State Department of Financial Services and the Federal Reserve Bank of New York, while the Los Angeles Agency is supervised by the California Department of Financial Protection and Innovation and the Federal Reserve Bank of San Francisco [R1].

Bank of Singapore, OCBC's private banking arm, has publicly described a Source of Wealth Assistant that uses agentic AI to automate part of the KYC due diligence process. The bank states that the tool reduces source-of-wealth report preparation from around 10 days to one

hour and that relationship managers review and refine the AI-generated drafts before internal AML/CFT review [R2].

This creates a strong RegTech design problem. The efficiency gain is valuable, but it also increases the need for governance because speed and narrative polish can be mistaken for evidentiary quality.

3. What the tool does and does not do

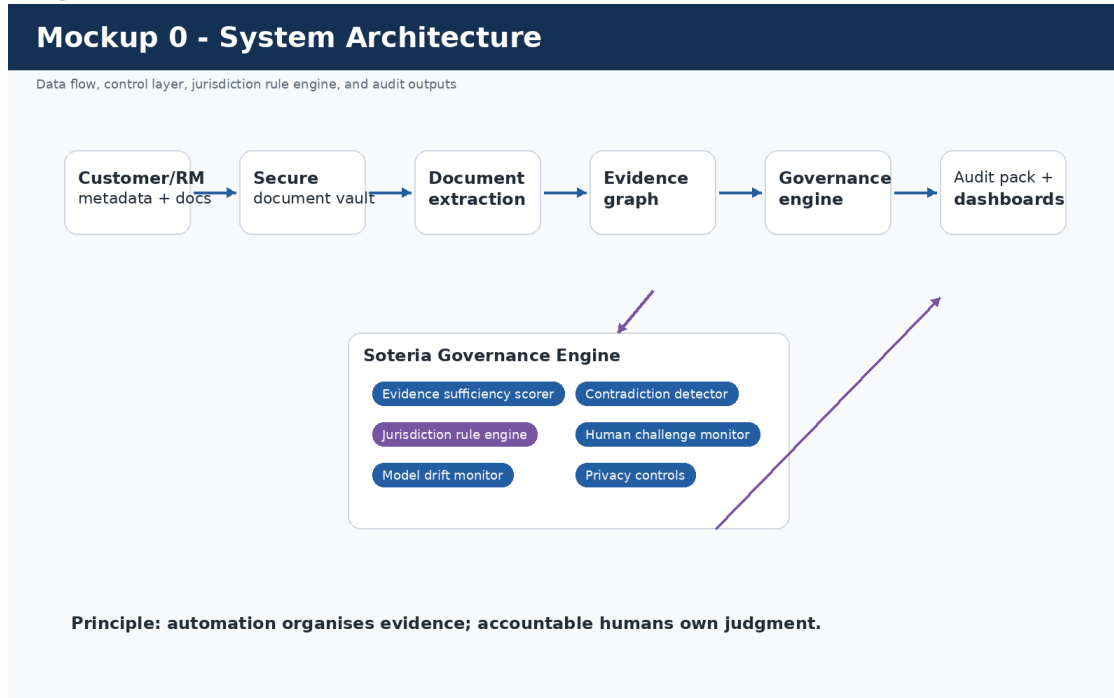
3.1 What the tool does

Layer	Function
Evidence layer	Links each material AI-generated wealth claim to supporting documents.
AI governance layer	Tracks unsupported claims, hallucination risk, contradiction rate, drift, override patterns, and human challenge.
Jurisdiction layer	Applies Singapore or US rule logic based on booking location, touchpoints, AI system type, and case type.
Human judgment layer	Forces named human review for red-friction cases, jurisdictional conflicts, and weak evidence.
Audit layer	Produces a governance pack showing rule version, evidence status, reviewer actions, exceptions, and unresolved risks.

3.2 What the tool does not do

Boundary	Reason
It does not approve or reject customers automatically.	Source-of-wealth conclusions require human compliance judgment.
It does not file STRs or SARs automatically.	Suspicious transaction/activity reporting decisions require legal and compliance judgment.
It does not claim that a case is simply “MAS compliant” or “US compliant.”	That would hide uncertainty and oversimplify regulatory judgment.
It does not perform full AML transaction monitoring.	Its scope is governance of AI-assisted KYC/source-of-wealth review.
It does not train general-purpose AI models on customer documents.	Source-of-wealth records are sensitive financial data.
It does not prove that wealth is legitimate.	It only assesses whether the bank’s conclusion is supported by available evidence and proper governance.

4. High-level architecture



System Architecture

4.1 System components

Component	Function
Secure document vault	Stores source-of-wealth documents with role-based access, access logs, and retention rules.
Document extraction layer	Extracts names, dates, entities, amounts, assets, jurisdictions, and ownership references.
Evidence graph	Links AI-generated claims to supporting documents and marks missing or partial support.
Evidence sufficiency scorer	Calculates evidence coverage, unsupported claims, and contradiction metrics.
Jurisdiction rule engine	Selects Singapore, US, or dual-rule controls and records rule version/effective date.
Human challenge workflow	Records whether reviewers amend, reject, escalate, or override AI outputs.
Model monitoring module	Tracks drift, extraction errors, unsupported claims, post-review escalations, and reviewer correction rates.
Audit pack generator	Produces a case-level governance record for

Component	Function
	compliance, internal audit, and senior management.

4.2 Data flow

1. Relationship manager creates source-of-wealth case
2. Case metadata identifies booking location, customer type, and cross-border touchpoints
3. Customer documents are uploaded into the secure document vault
4. The bank's AI assistant drafts the source-of-wealth narrative
5. Soteria extracts material claims from the AI draft
6. Evidence graph links each claim to documents or marks it unsupported
7. Jurisdiction rule engine activates Singapore, US, or dual-rule controls
8. Evidence sufficiency and contradiction scores are calculated
9. Human review workflow assigns required reviewers and escalation level
10. Audit pack records rule version, evidence map, review actions, and unresolved risks

5. Synthetic data used in the design

The repository includes a synthetic dataset at data/synthetic_sow_cases.csv. It contains 30 simulated source-of-wealth governance cases. The dataset is included to show that the architecture has specific inputs and output logic rather than being only a conceptual diagram.

5.1 Dataset fields

Field	Purpose
case_id	Synthetic case identifier.
jurisdiction	Singapore, United States, or Cross-border.
booking_location	Booking location or agency context.
sg_touchpoint / us_touchpoint	Flags whether Singapore or US controls may be relevant.
customer_type	Simulated customer category.
ai_system_type	Agentic AI, generative AI, non-generative AI, or traditional statistical model.
evidence_coverage_ratio	Supported material claims divided by total material claims.
unsupported_claim_rate	Unsupported AI material claims divided by total material AI claims.
contradiction_count	Number of material conflicts between the AI narrative and source documents.
human_challenge_rate	Indicator of whether reviewers are challenging AI outputs.

Field	Purpose
model_drift_flag	Whether model drift controls are triggered.
jurisdiction_conflict_flag	Whether dual-rule review is required.
privacy_exception_flag	Whether data-handling controls are triggered.
final_tool_output	Green, Amber, Red Friction, Amber + Governance Gap, or Red Friction + Governance Gap.

5.2 Data limitation

The data is synthetic because real KYC/source-of-wealth files are confidential, sensitive, and unsuitable for a student assignment. The dataset is not calibrated against actual OCBC or Bank of Singapore customer outcomes.

6. Scoring methodology

6.1 Core formulas

Evidence Coverage Ratio = supported material claims / total material claims

Unsupported Claim Rate = unsupported AI-generated material claims / total AI-generated material claims

Contradiction Count = number of material conflicts between AI narrative and source documents

Human Challenge Rate = cases where a human amended, rejected, or escalated AI output / total AI-assisted cases

Override Quality Score = percentage of overrides with case-specific reasons rather than generic approval text

6.2 Thresholds

Condition	Output
Evidence coverage $\geq 85\%$, unsupported claim rate $\leq 10\%$, contradiction count = 0, no governance gap	Green
Evidence coverage 70-84%, unsupported claim rate 11-20%, contradiction count ≤ 1	Amber
Evidence coverage $< 70\%$, unsupported claim rate $> 20\%$, or contradiction count ≥ 2	Red Friction
US case + generative AI or agentic AI	Add Governance Gap Warning

Condition	Output
Cross-border case with Singapore and US touchpoints	Activate Dual-Rule Review
Model drift flag = true	Red Friction and model-risk review
Human challenge rate below 20%	Human-overreliance warning

A case cannot receive a Green output if material contradictions remain unresolved, even if all mandatory documentation fields are complete.

7. Decision points and owners

Decision point	Question asked	Automated or human?	Owner	Output
DP1 Jurisdiction selection	Is this case Singapore, US, or cross-border?	Automated with human confirmation	Compliance Operations	Active rule set
DP2 AI system classification	Is the system traditional, non-generative AI, generative AI, or agentic AI?	Human-approved	Model Risk Manager	Model governance category
DP3 Evidence sufficiency	Are material source-of-wealth claims supported?	Automated scoring plus human review	Financial Crime Compliance	Green / Amber / Red Friction
DP4 Contradiction detection	Do source documents conflict with the AI narrative?	Automated flagging plus human judgment	Compliance Officer	Escalation if material
DP5 AML/KYC risk level	Is enhanced due diligence required?	Automated trigger plus compliance decision	Financial Crime Compliance	Standard review / EDD / escalation
DP6 Human challenge	Did the reviewer meaningfully challenge the AI output?	Automated monitoring plus audit review	Compliance Head / Internal Audit	Challenge quality score
DP7 Final source-of-wealth approval	Can the source-of-wealth conclusion be accepted?	Human only	Senior Compliance Officer	Approve / reject / request more evidence
DP8 Rule update approval	Should a new rule version be	Human only	Legal/ Compliance rule	Updated rule version

Decision point	Question asked	Automated or human?	Owner	Output
	released?		owner	
DP9 Model drift response	Should use be paused or revalidated?	Human decision after system alert	Model Risk Committee	Revalidation / threshold change / freeze
DP10 STR/SAR decision	Is suspicious reporting required?	Human only	Financial Crime Compliance / Legal	File / no file / further review

The design principle is that automation can organise evidence, but it cannot own judgment.

8. Jurisdiction rule engine

The jurisdiction rule engine is the mechanism that makes the tool genuinely jurisdiction-aware. It stores rules in a versioned repository with rule ID, jurisdiction, source, domain, trigger, control action, effective date, version, owner, and status.

8.1 Rule selection decision tree

Start

```

|
|-- Is booking location Singapore or is there a Singapore touchpoint?
|   |-- Yes: Activate SG-AML-626 + SG-AI-MRM + SG-FEAT/Veritas
|   |-- No: Continue
|
|-- Is booking location United States or is a US agency/account involved?
|   |-- Yes: Activate US-CDD-FINCEN
|   |-- No: Continue
|
|-- What type of AI system is used?
|   |-- Traditional/statistical or non-generative AI + US: apply US-MRM-SR26-2
|   |-- Generative/agentive AI + Singapore: apply MAS AI governance controls
|   |-- Generative/agentive AI + US: issue US-AI-GAP governance warning
|
|-- Are both Singapore and US touchpoints present?
|   |-- Yes: Activate dual-rule review and no-silent-downgrade rule
|   |-- No: Continue single-jurisdiction review
|
|-- Calculate evidence and contradiction scores
|
|-- Apply Green / Amber / Red Friction output logic

```

8.2 Rule update and versioning process

Step	Process
Source detection	Regulatory update detected from MAS, FinCEN, Federal Reserve, OCC, state regulator, or internal policy.
Legal/compliance review	Human owner confirms whether update applies to the relevant business line.
Rule drafting	Rule engineer converts the change into structured rule logic.
Dual approval	Compliance owner and technology owner approve the rule.
Version release	Rule is published with version number, effective date, expiry date if applicable, and owner.
Open-case impact analysis	System identifies open cases affected by the change.
Re-run or grandfathering decision	Compliance decides whether open cases must be reassessed.
Audit record	Old rule, new rule, approvers, affected cases, and decisions are logged.

9. Jurisdiction-specific regulatory logic

9.1 Singapore mode

In Singapore mode, the tool treats AI-assisted source-of-wealth review as both an AML/KYC issue and an AI governance issue. MAS has published an AI Model Risk Management information paper based on a thematic review of banks' AI, including generative AI model risk management practices [R3]. MAS also has FEAT principles and the Veritas initiative for responsible AI and data analytics evaluation [R4, R5]. MAS Notice 626 sets AML/CFT requirements for banks, including risk assessment, risk mitigation, and customer due diligence [R6].

Singapore mode activates the following controls:

Control area	Singapore-mode control
AI inventory	Record the AI system, owner, intended use, model type, and limits.
Human oversight	Require human review for all material source-of-wealth conclusions.
Evidence quality	Link every material wealth claim to supporting evidence or mark it unsupported.
FEAT/Veritas	Check accountability, transparency to

Control area	Singapore-mode control
AML/KYC	internal users, fairness monitoring, and reviewer ownership. Apply customer due diligence, enhanced due diligence triggers, source-of-wealth evidence requirements, and ongoing monitoring logic.
Audit pack	Produce MAS-facing governance record showing rule version, evidence map, human review, and exceptions.

9.2 US mode

In US mode, the tool applies US customer due diligence and model-risk logic. FinCEN's CDD materials describe customer due diligence obligations for covered financial institutions, including customer identification, beneficial ownership when required, customer risk profile, and ongoing monitoring [R7].

For model risk, the 2026 Federal Reserve and OCC materials distinguish traditional/non-generative model-risk governance from generative and agentic AI. The Federal Reserve guidance says generative AI and agentic AI are not within the scope of that guidance, while broader risk management and governance practices should guide controls for tools not covered by the guidance [R8]. OCC Bulletin 2026-13 states a similar exclusion for generative and agentic AI models [R9].

US mode activates the following controls:

Control area	US-mode control
CDD	Customer identification, beneficial ownership when required, relationship purpose, risk profile, and ongoing monitoring.
Model risk	Apply SR 26-2-style controls only where the tool is a traditional/statistical or non-generative/non-agentic model.
AI governance gap	For generative or agentic AI, issue a governance gap warning rather than falsely marking the system as fully covered by SR 26-2.
Human approval	Require human compliance approval before accepting high-risk source-of-wealth conclusions.
Branch/agency context	Record whether the US touchpoint is New York Agency, Los Angeles Agency, or another US relationship.
Audit pack	Produce US-facing governance record

Control area	US-mode control
	showing CDD controls, AI governance gap, rule version, and human review.

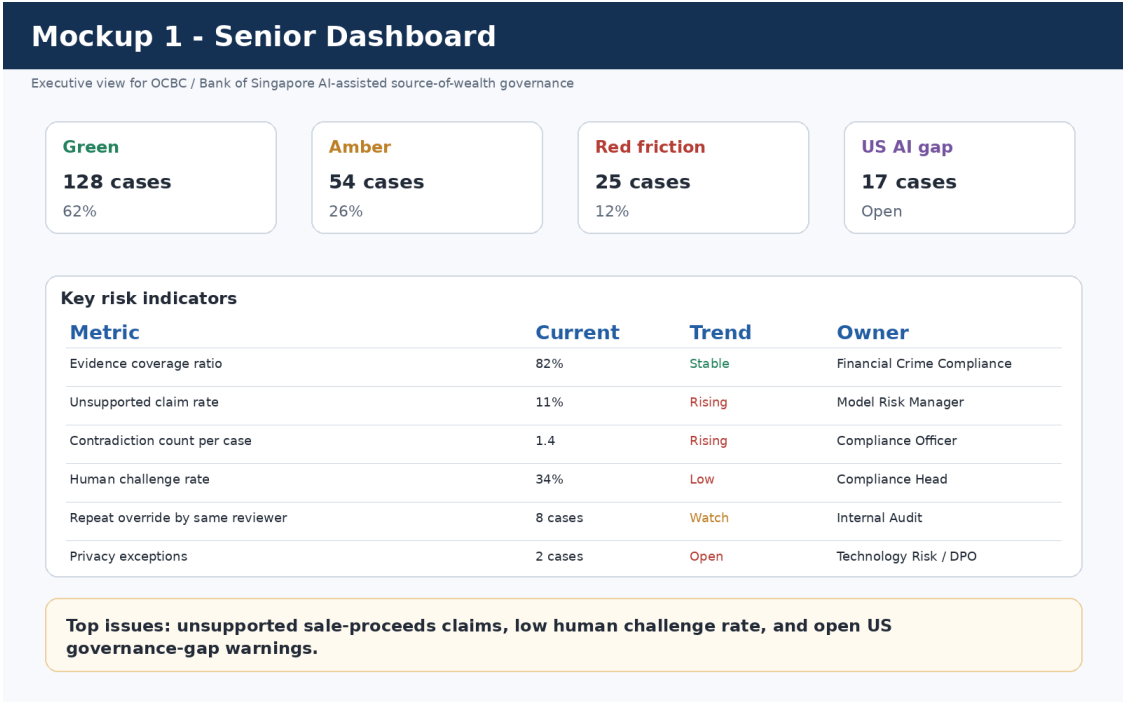
9.3 Same case, different jurisdictional output

Same case fact	Singapore output	US output
Agentic AI used in KYC source-of-wealth workflow	MAS AI MRM, FEAT, and Veritas-style controls triggered.	Governance gap warning because generative/agentic AI is outside the specific 2026 model-risk guidance scope.
Source-of-wealth evidence incomplete	MAS Notice 626 CDD/EDD review triggered.	FinCEN CDD/customer risk profile review triggered.
Human reviewer approves without edits	Human accountability weakness flagged.	Effective challenge/internal governance weakness flagged.
Contradictions in sale proceeds	Red Friction and senior compliance review.	Red Friction + CDD escalation; add governance gap if AI is agentic/generative.
Singapore and US touchpoints both present	Dual-rule review and no-silent-downgrade rule.	Dual-rule review and no-silent-downgrade rule.

This table is the core jurisdiction-aware design feature. The same operational facts produce different regulatory explanations depending on jurisdiction.

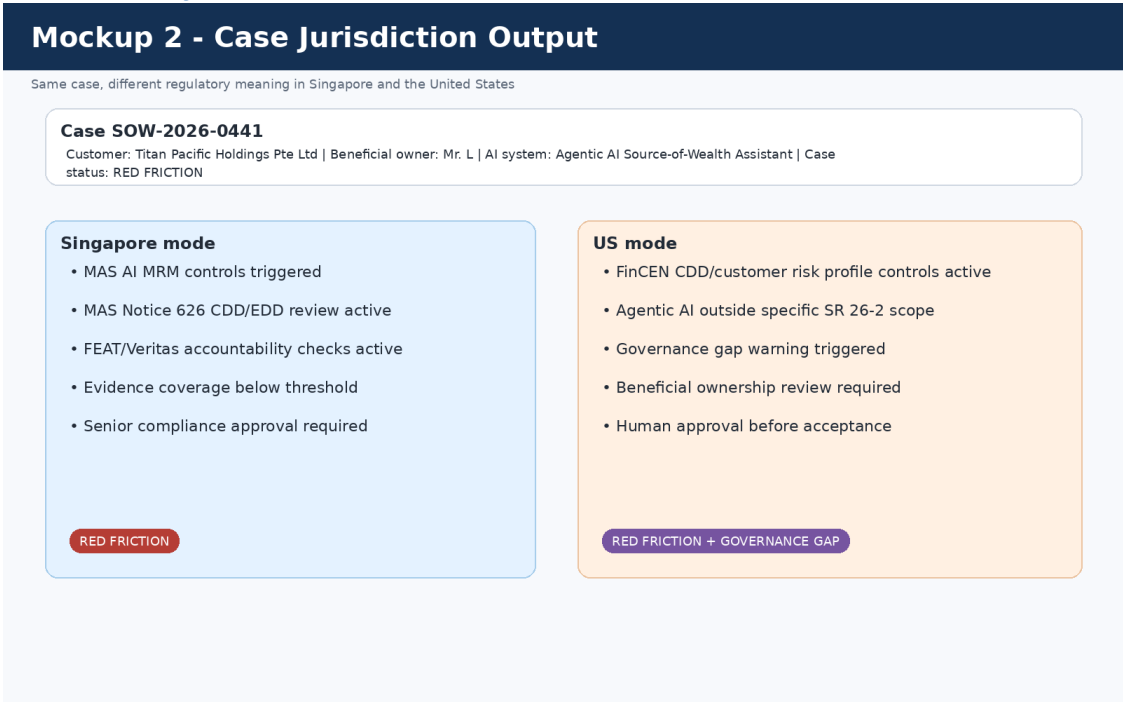
10. Mockups

10.1 Senior dashboard



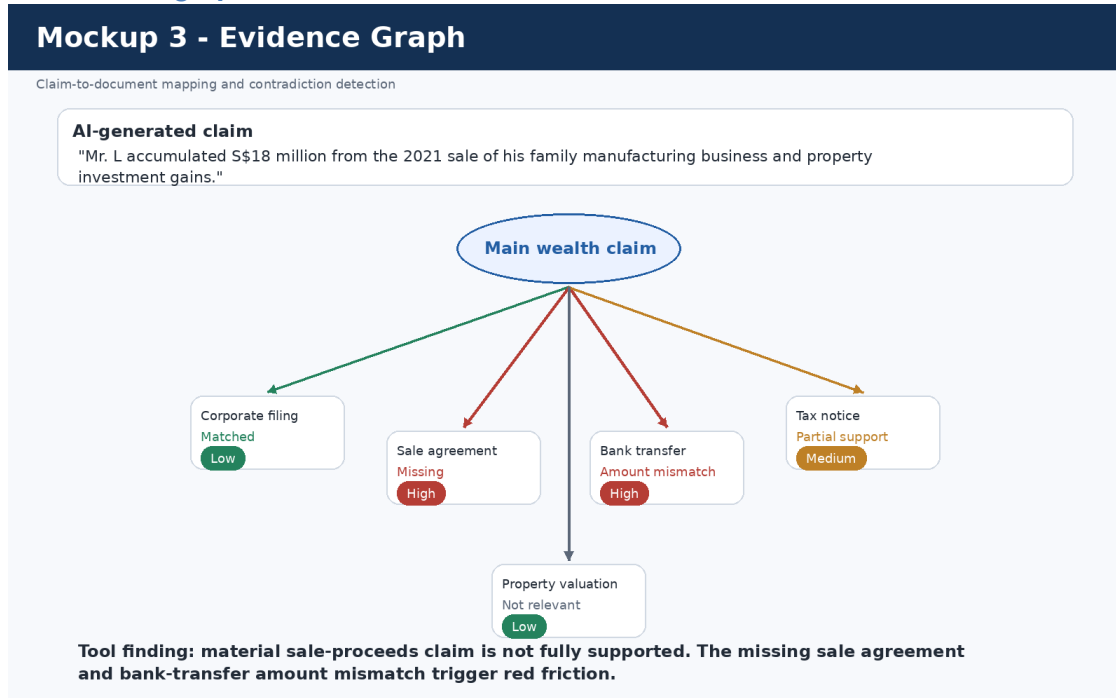
Senior Dashboard

10.2 Case-level jurisdiction output



Case Jurisdiction Output

10.3 Evidence graph



Evidence Graph

10.4 Human challenge log

Mockup 4 - Human Challenge Log

Testing whether human-in-the-loop review is meaningful

Reviewer actions for Case SOW-2026-0441
Reviewer: Relationship Manager A
AI recommendation: Accept source-of-wealth narrative

Tool status: RED FRICTION

- ☐ Confirm sale agreement obtained
- ☒ Explain amount mismatch
- ☒ Confirm beneficial owner identity
- ☐ Explain why missing evidence is acceptable
- ☒ Escalate to Financial Crime Compliance
- ☐ Senior compliance review completed

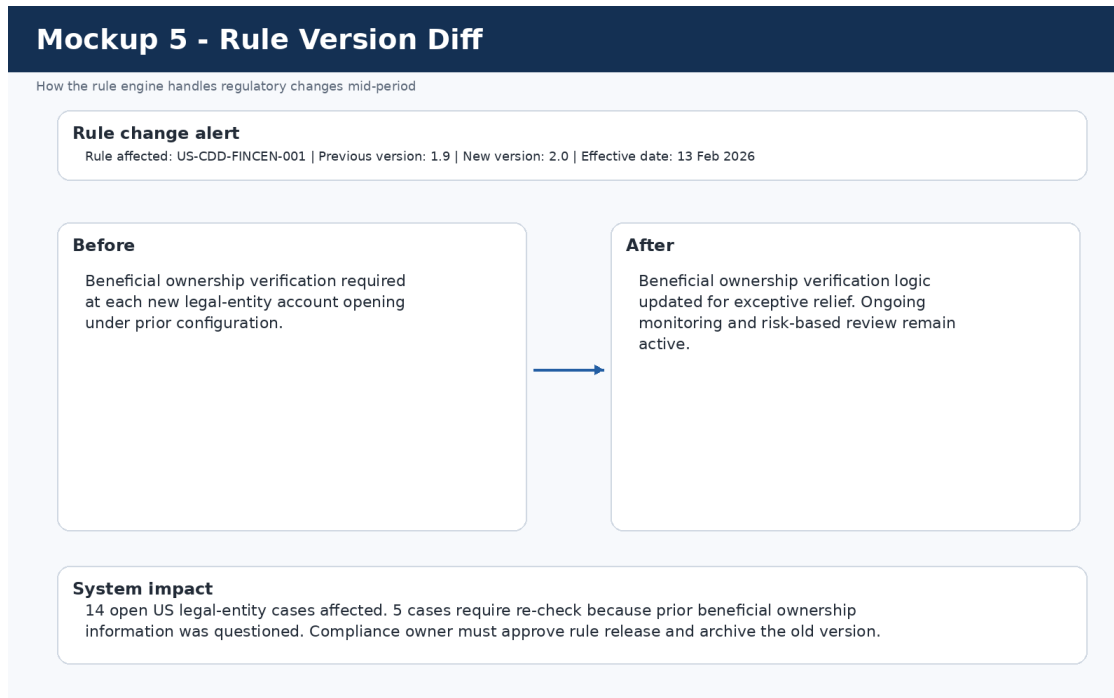
Override rationale

No override allowed yet. The sale agreement is missing and the bank transfer amount does not match the AI narrative. Senior compliance review is required before the source-of-wealth conclusion can be accepted.

Decision: Pending senior review

Human Challenge Log

10.5 Rule-version diff screen



Rule Version Diff

11. Failure-mode analysis

Failure mode	Cause	Impact	Likelihood	Severity	Control	Owner
Rule changes mid-period	Rule engine not updated or open cases not re-run	Cases assessed under obsolete requirements	Medium	High	Versioned rules, effective dates, rule-diff alert, open-case impact analysis	Compliance rule owner
Model drift	AI behaviour, typologies, or document formats change	More hallucinated or unsupported source-of-wealth claims	Medium	High	Drift thresholds, unsupported-claim monitoring, model-risk review	Model Risk Manager
Human rubber-stamping	Reviewers approve AI drafts	Human-in-the-loop becomes	High	High	Human Challenge Log and	Financial Crime Compliance

Failure mode	Cause	Impact	Likelihood	Severity	Control	Owner
	without challenge	symbolic			reviewer-pattern monitoring	e Head
Jurisdictional misconfiguration	Wrong booking location or cross-border flag selected	Wrong rule set applied	Medium	High	Booking-location check, branch/agency check, dual-rule activation	Compliance Operations
Contradictory jurisdictional requirements	US and Singapore controls point in different directions	Confusion over whether stricter or lighter controls apply	Medium	High	No-silent-downgrade rule and additive controls	AI Governance Committee
Privacy/data-security failure	Sensitive documents processed or retained improperly	Customer privacy harm and regulatory exposure	Low/Medium	Very High	Secure vault, role-based access, access logs, retention controls	Technology Risk / DPO
False negative	Illicit or unsupported wealth accepted as legitimate	Financial crime risk enters the banking system	Medium	Very High	Red Friction, senior compliance approval, contradiction detection	Financial Crime Compliance
False positive	Legitimate customer incorrectly flagged	Delayed onboarding and relationship harm	Medium	Medium	Explainable escalation reason and proportional follow-up	Relationship Manager + Compliance Officer
Threshold miscalibration	Thresholds tuned for efficiency	Too many weak cases pass or too	Medium	High	Periodic threshold review,	Model Risk Manager

Failure mode	Cause	Impact	Likelihood	Severity	Control	Owner
	rather than risk	many good cases escalate			sensitivity testing, outcome monitoring	
Vendor interface design failure	Dashboard hides uncertainty or over-emphasizes green status	Users gain false confidence	Medium	High	No simple compliance badge, uncertainty labels, audit logs	Soteria Product Owner

11.1 What happens when a rule changes mid-period?

The rule engine stores effective dates and version numbers. When a rule changes, the tool creates a rule-diff alert, identifies affected open cases, and requires a compliance owner to decide whether cases should be re-run or grandfathered. The old and new rules are both retained for audit.

11.2 What happens when a model drifts?

The tool monitors unsupported claim rate, contradiction count, reviewer correction rate, post-review escalation rate, and extraction error rate. If drift crosses threshold, the tool freezes straight-through approval, requires enhanced human review, notifies model risk management, and produces a drift incident report.

11.3 What happens when jurisdictions contradict each other?

The tool applies a no-silent-downgrade rule. If one jurisdiction creates a governance gap and another jurisdiction requires active governance, the tool does not treat the gap as permission to reduce controls. Cross-border cases receive additive controls until a human legal/compliance owner approves a jurisdictional interpretation.

12. Privacy and data-governance architecture

Source-of-wealth review involves highly sensitive financial and personal data, including tax records, bank statements, payslips, corporate filings, beneficial ownership details, investment records, and property documents. Privacy is therefore a core design requirement.

Control	Description
Secure document vault	Raw documents are stored in a controlled bank environment.
Role-based access	Access is limited to authorized relationship managers, compliance reviewers, audit, and

Control	Description
Access logs	model-risk users. All document access and evidence-map changes are logged.
No uncontrolled model training	Customer documents are not used to train general-purpose models.
Retention controls	Raw documents are retained only under approved policies.
Cross-border transfer flag	Cases involving US and Singapore touchpoints are flagged for data-transfer review.
Separation of pipelines	Evidence extraction is separated from any model-improvement process.

13. Implementation-lite decision logic

A simplified version of the rule logic is included in `rule_engine/pseudocode_decision_logic.md`. The following simplified logic shows the core approach:

```

if singapore_touchpoint:
    activate(["SG-AML-626", "SG-AI-MRM", "SG-FEAT"])

if us_touchpoint:
    activate(["US-CDD-FINCEN"])

if ai_system_type in ["generative_ai", "agentic_ai"] and us_touchpoint:
    activate(["US-AI-GAP"])
    warn("Do not mark as fully covered by SR 26-2")

if evidence_coverage_ratio < 0.70 or unsupported_claim_rate > 0.20 or
contradiction_count >= 2:
    output = "RED_FRICTION"
elif "US-AI-GAP" in active_rules:
    output = "AMBER_WITH_GOVERNANCE_GAP"
else:
    output = "GREEN_OR_AMBER_BASED_ON_EVIDENCE_SCORE"

```

14. What I did not build because of time and data limits

1. I did not calibrate thresholds using real OCBC or Bank of Singapore customer files.
2. I did not validate model drift against production AI outputs.

3. I did not test the tool against real suspicious transaction or suspicious activity reporting outcomes.
4. I did not build a live integration with OCBC systems, MAS systems, FinCEN systems, or screening vendors.
5. I did not provide legal advice on MAS, FinCEN, Federal Reserve, OCC, or state regulator obligations.
6. I used synthetic data because real KYC/source-of-wealth data is confidential and sensitive.
7. I did not build a full AML transaction monitoring system; this tool is limited to governance of AI-assisted source-of-wealth review.

These limitations are important. The tool is a defensible architecture and mockup, not a production compliance system.

15. Why a memo or spreadsheet cannot solve this problem

Problem	Memo/spreadsheet limitation	Tool advantage
Unsupported AI claims	Manual reviewer may miss unsupported narrative claims.	Tool links every material claim to documents and flags missing evidence.
Contradictions	Hard to compare many documents manually.	Tool compares dates, amounts, names, entities, and jurisdictions.
Jurisdiction rules	Spreadsheet rules become outdated.	Rule engine stores versions, effective dates, and owners.
Model drift	Hard to detect over time.	Tool monitors unsupported claims, contradictions, and review outcomes.
Human rubber-stamping	Memo records approval, not quality of challenge.	Human Challenge Log tracks edits, overrides, and escalation behaviour.
Audit readiness	Evidence is fragmented.	Audit pack records rule version, evidence map, reviewer actions, and exceptions.

16. Final design position

My final design position is that AI can assist source-of-wealth review, but it should not be allowed to convert uncertainty into false confidence. The tool therefore has four commitments:

1. **Evidence first:** every material AI claim should be linked to support or marked unsupported.

2. **Human accountability:** final source-of-wealth judgment must remain with identifiable human reviewers.
 3. **Jurisdictional honesty:** Singapore and US controls must be applied differently where the regulatory logic differs.
 4. **No cosmetic compliance:** the tool should not issue a simple “compliant” badge when evidence gaps, model drift, privacy issues, or governance gaps remain unresolved.
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